

Statistics and probability

Course code –ECE-520809

Introduction & Meaning of Statistics: meaning of these words is “**Political State**” or a Government. Shakespeare used a word Statist in his drama Hamlet (1602). In the past, the statistics was used by rulers. The application of statistics was very limited but rulers and kings needed information about lands, agriculture, commerce, population of their states to assess their military potential, their wealth, taxation and other aspects of government.

Gottfried Achenwall used the word statistics at a German University in 1749 which means that political science of different countries. In 1771 W. Hooper (Englishman) used the word statistics in his translation of Elements of Universal Erudition written by Baron B.F Bieford, in his book statistics has been defined as the science that teaches us what is the political arrangement of all the modern states of th

The Word statistics have been derived from Latin word “**Status**” or the Italian word “**Statista** e known world. There is a big gap between the old statistics and the modern statistics, but old statistics also used as a part of the present statistics. During the 18th century the English writer have used the word statistics in their works, so statistics has developed gradually during last few centuries. A lot of work has been done in the end of the nineteenth century.

At the beginning of the 20th century, William S Gosset was developed the methods for decision making based on small set of data. During the 20th century several statistician are active in developing new methods, theories and application of statistics. Now these days the availability of electronics computers is certainly a major factor in the modern development of statistics.

Statistics is the science of data processing. It is a branch of knowledge dealing with collection, organization, presentation, analysis and interpretation of numerical data. Primarily statistics has two meanings. Statistics in plural sense refers to mass of information relating to characteristics of a population which can be defined as the aggregates of units under study. For example – the amount of recent per capita (average) expenditure of the official’s workers of an industry, the annual birth rate or death rate of Bangladesh etc.

Statistics in singular sense refers to the science of statistics which performs statistical functions dealing with data collected for the purpose of specific study of a problem.

Definition of Statistics

The statisticians have defined statistics in different ways According to Prof. A.L. Bowley gives the three definitions of Statistics which as follows:

1. Statistics may be called the science of counting. This definition is too narrow because it covers only one aspect of the science, namely collection of data.
2. Statistics may be called the science of averages. This definition also is not satisfactory because averages are not only one of the devices used in statistical analysis.
3. Statistics is the science of the measurement of social organism, regarded as a whole in all its manifestations. This definition also is not satisfactory because it confines the scope of statistics only to sociology.

Boddington defines statistics as “the science of estimates and probabilities.” This definition also is unacceptable because estimates and probabilities are only a part of statistical method.

Croxton and Cowden say, “Statistics may be defined as a science of collection, presentation, analysis and interpretation of numerical data.” This definition clearly points out four stages in statistical investigation, namely: (1) collection of data (2) presentation of data (3) analysis of data and (4) interpretation of data.

However, to the above stages one more stage may be added and that is the organization of data. Thus, **statistics may be defined as the science of collection, organization, presentation, analysis and interpretation of numerical data.**

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data. In applying statistics to, e.g., a scientific, industrial, or social problem, it is conventional to begin with a statistical population or a statistical model process to be studied. Populations can be diverse topics such as “all people living in a country” or “every atom composing a crystal”. Statistics deals with all aspects of data including the planning of data collection in terms of the design of surveys and experiments.

Statistics is a form of mathematical analysis that uses quantified models, representations and synopses for a given set of experimental data or real-life studies. Statistics studies methodologies to gather, review, analyze and draw conclusions from data. Some statistical measures include central value measurement, regression analysis, variance analysis, time series analysis, probability measurement, sampling, hypothesis testing etc.

Statistics is the mathematical science involved in the application of quantitative principles to the collection, analysis, and presentation of numerical data. The practice of statistics utilizes data from some population in order to describe it meaningfully, to draw conclusions from it, and make informed decisions. The population may be a community, an organization, a production line, a service counter, or a phenomenon such as the weather. Statisticians determine which quantitative model is correct for a given type of problem and they decide what kinds of data should be collected and examined. Applied statistics concerns the application of the general methodology to particular problems. This often calls for use of the techniques of computer-based data analysis.

Statistical Method:

Statistical methods are the methods of collecting, summarizing, analyzing, and interpreting variable numerical data. Statistical methods can be contrasted with deterministic methods,

which are appropriate where observations are exactly reproducible or are assumed to be so. While statistical methods are widely used in the life sciences, in economics, and in agricultural science, they also have an important role in the physical sciences in the study of measurement errors, of random phenomena such as radioactivity or meteorological events, and in obtaining approximate results where deterministic solutions are hard to apply.

Data collection involves deciding what to observe in order to obtain information relevant to the questions whose answers are required, and then making the observations. Sampling involves choice of a sufficient number of observations representing an appropriate population. Experiments with variable outcomes should be conducted according to principles of experimental design.

Data summarization is the calculation of appropriate statistics and the display of such information in the form of tables, graphs, or charts. Data may also be adjusted to make different samples more comparable, using ratios, compensating factors, etc.

Statistical analysis relates observed statistical data to theoretical models, such as probability distributions or models used in regression analysis. By estimating parameters in the proposed model and testing hypotheses about rival models, one can assess the value of the information collected and the extent to which the information can be applied to similar situations. *Statistical prediction* is the application of the model thought to be most appropriate, using the estimated values of the parameters.

Finally we can say that **statistical methods are mathematical formulas, models, and techniques** that are used in statistical analysis of raw research data. The application of statistical methods extracts information from research data and provides different ways to assess the robustness of research outputs.

Objective of Statistics:

The main purpose of statistics is the collection of data regarding different aspects of life, presentation, analysis and coming to the conclusion with a reasonable interpretation. In this respect, the objectives of statistics are as follows:

- (1). Statistics collects the numerical and analytical data regarding various aspects of our daily life.
- (2). It presents the collected data attractively and simplifies them. For this classification, tabulation and graph-diagram are used.
- (3). Analyzing the mathematical data it presents them to us in compact form.
- (4). It deals with the characteristics of obtained information and compares among them.
- (5). It deals with various methods of experiment.

Functions or Uses of Statistics

Statistics helps in providing a better understanding and exact description of a phenomenon of nature.

- (1) Statistical helps in proper and efficient planning of a statistical inquiry in any field of study.
- (2) Statistical helps in collecting an appropriate quantitative data.
- (3) Statistics helps in presenting complex data in a suitable tabular, diagrammatic and graphic form for an easy and clear comprehension of the data.
- (4) Statistics helps in understanding the nature and pattern of variability of a phenomenon through quantitative observations.
- (5) Statistics helps in drawing valid inference, along with a measure of their reliability about the population parameters from the sample data.
- (6) Statistics helps in providing a better understanding and exact description of a phenomenon of nature.
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- (11) Statistical helps in collecting an appropriate quantitative data.
- (12) Statistics helps in presenting complex data in a suitable tabular, diagrammatic and graphic form for an easy and clear comprehension of the data.
- (13) Statistics helps in understanding the nature and pattern of variability of a phenomenon through quantitative observations.
- (14) Statistics helps in drawing valid inference, along with a measure of their reliability about the population parameters from the sample data.

Types of Statistics:

- (i) Descriptive Statistics
- (ii) Inferential or Inductive Statistics

Descriptive Statistics & Inferential or Inductive Statistics:

Descriptive & Inductive Statistics refer to the application of statistical methods & theories. Descriptive Statistics deals with collection, presentation, analysis & interpretation of statistical data without involving generalizations. On the other hand, generalization, predictions, estimations & decisions made from collected & analyzed data using statistical methods & theories come within the fold of Inductive Statistics. When we compute average with the sole purpose of describing a given phenomenon, we are using the mean as a tool of descriptive statistics, whereas if we make use of the average for making prediction or general conclusion, we are using it in inductive sense.

For example: Average salary of the employees of a factory may be described by calculating the mean salary of all the employees of the factory. This involves only description of mean salary. But if this mean averages of a particular factory is used as an estimate of average salary of the

employees of all the factories in an industry instead of computing the average for all the employees in the industry then we use the mean in inductive sense.

Scope and Importance of Statistics:

1. **Statistics and planning:** Statistics is indispensable in planning in the modern age which is termed as "the age of planning". Almost all over the world the govt. are resorting to planning for economic development.
2. **Statistics and mathematics:** Statistics are intimately related recent advancements in statistical technique are the outcome of wide applications of mathematics.
3. **Statistics and modern science:** In medical science the statistical tools for collection, presentation and analysis of observed facts relating to causes and incidence of diseases and the result of application various drugs and medicine are of great importance.
4. **Statistics and communication Engineering:** In **communication engineering** there are several instances where **statistical** modeling is **used** for **system** planning, design, operation, and management. ... **Statistical** analysis and modeling of a network gives us information about the network deployment costs, network dynamics, and parameters of network traffic.
5. **Statistics and computer science:** **Statistics** is **used** for data mining, speech recognition, vision and image analysis, data compression, artificial intelligence, and network and traffic modeling. A **statistical** background is essential for understanding algorithms and **statistical** properties that form the backbone of **computer science**.
6. **Statistics and economics:** Statistical data and techniques of statistical analysis have to immensely useful involving economical problem. Such as wages, price, time series analysis, demand analysis.
7. **Statistics and business:** Statistics is an irreplaceable tool of production control. Business executives are relying more and more on statistical techniques for studying the needs and desires of the valued customers.
 - i. **Marketing:** Statistical analysis is frequently used in providing information for making decision in the field of marketing. It is necessary first to find out what can be sold and then to evolve suitable strategy, so that the goods can be taken by the ultimate consumer. A skill full analysis of data on production purchasing power, man power, habits of competitors, habits of consumer, transportation cost should be considered to take any attempt to establish a new market.
 - ii. **Production:** In the field of production statistical data and method play a very important role. The decision about what to produce? How to produce? When to produce? For whom to produce is based largely on statistical analysis.
 - iii. **Finance:** The financial organization discharging their finance function effectively depend very heavily on statistical analysis of profit and losses.
 - iv. **Banking:** Banking institutions have found it increasingly to establish research department within their organization for the purpose of gathering and analysis information, not only regarding their

own business but also regarding general economic situation and every segment of business in which they may have interest.

v. Investment: Statistics greatly assists investors in making clear and valued judgment in his investment decision in selecting securities which are safe and have the best prospects of yielding a good income.

vi. Purchase: The purchase department in discharging their function makes use of statistical data to frame suitable purchase policies such as what to buy? What quantity to buy? What time to buy? Where to buy? Whom to buy?

vii. Accounting: Statistical data are also employer in accounting particularly in auditing function, the technique of sampling and destination is frequently used.

viii. Control: The management control process combines statistical and accounting method in making the overall budget for the coming year including sales, materials, labor and other costs and net profits and capital requirement.

8. Statistics and industry: In industry statistics is widely used inequality control. In production engineering to find out whether the product is confirming to the specifications or not. Statistical tools, such as inspection plan, control chart etc.

9. Statistics and psychology & education: In education and physiology statistics has found wide application such as, determining or to determine the reliability and validity to a test, factor analysis etc.

10. Statistics and war: In war the theory of decision function can be a great assistance to the military and personal to plan "maximum destruction with minimum effort."

Limitations of Statistics

Statistics is indispensable to almost all sciences - social, physical and natural. It is very often used in most of the spheres of human activity. In spite of the wide scope of the subject it has certain limitations. Some important limitations of statistics are the following:

- 1. Statistics does not study qualitative phenomena:** Statistics deals with facts and figures. So the quality aspect of a variable or the subjective phenomenon falls out of the scope of statistics. For example, qualities like beauty, honesty, intelligence etc. cannot be numerically expressed. So these characteristics cannot be examined statistically. This limits the scope of the subject.

2. **Statistical laws are not exact:** Statistical laws are not exact as in case of natural sciences. These laws are true only on average. They hold good under certain conditions. They cannot be universally applied. So statistics has less practical utility.
 3. **Statistics does not study individuals:** Statistics deals with aggregate of facts. Single or isolated figures are not statistics. This is considered to be a major handicap of statistics.
 4. **Statistics can be misused:** Statistics is mostly a tool of analysis. Statistical techniques are used to analyze and interpret the collected information in an enquiry. As it is, statistics does not prove or disprove anything. It is just a means to an end. Statements supported by statistics are more appealing and are commonly believed. For this, statistics is often misused. Statistical methods rightly used are beneficial but if misused these become harmful. Statistical methods used by less expert hands will lead to inaccurate results. Here the fault does not lie with the subject of statistics but with the person who makes wrong use of it.
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Questions

1. What is statistics? Discuss the importance of statistics in Computer science.
2. Discuss applications and limitations of statistics.
3. Explain some important functions of statistics.
4. How does statistics help a programmer?
5. 'Statistics is the science of average' – Explain.
6. What is the basic difference between Descriptive and Inferential Statistics?

Collection, Proccession & Presentation of Data

The description of statistical data may be quite elaborate or quite brief depending on two factors: the nature of data and the purpose for which the same data have been collected. While describing data statistically or verbally, one must ensure that the description is neither too brief nor too lengthy. The measures of central tendency enable us to compare two or more distributions pertaining to the same time period or within the same distribution over time. For example, the average consumption of tea in two different territories for the same period or in a territory for two years, say, 2015 and 2016, can be attempted by means of an average.

The two types of data are grouped and ungrouped. Grouped data are data organized into a frequency distribution. Differentiating between grouped and ungrouped data is important, because statistical operations on the two types are computed differently. Constructing a frequency distribution involves several steps. The first step is to determine the range of the data, which is the difference between the largest value and the smallest value. Next, the number of classes is determined, which is an arbitrary choice of the researcher. However, too few classes over-aggregate the data into meaningless categories, and too many classes do not summarize the data enough to be useful. The third step in constructing the frequency distribution is to determine the width of the class interval. Dividing the range of values by the number of classes yields the approximate width of the class interval. The class midpoint is the midpoint of a class interval. It is the average of the class endpoints and represents the halfway point of the class interval. Relative frequency is a value computed by dividing an individual frequency by the sum of the frequencies. Relative frequency represents the proportion of total values that is in a given class interval. The cumulative frequency is a running total frequency tally that starts with the first frequency value and adds each ensuing frequency to the total.

Two types of graphical depictions are quantitative data graphs and qualitative data graphs. Quantitative data graphs presented in this chapter are histogram, frequency polygon, ogive, dot plot, and stem-and-leaf plot. Qualitative data graphs presented are pie chart, bar chart, and Pareto chart. In addition, two-dimensional scatter plots are presented. A histogram is a vertical bar chart in which a line segment connects class endpoints at the value of the frequency. Two vertical lines connect this line segment down to the x-axis, forming a rectangle. A frequency polygon is constructed by plotting a dot at the midpoint of each class interval for the value of each frequency and then connecting the dots. Ogives are cumulative frequency polygons. Points on an ogive are plotted at the class endpoints. A dot plot is a graph that displays frequency counts for various data points as dots graphed above the data point. Dot plots are especially useful for observing the overall shape of the distribution and determining both gaps in the data and high concentrations of data.

Stem-and-leaf plots are another way to organize data. The numbers are divided into two parts, a stem and a leaf. The stems are the leftmost digits of the numbers and the leaves are the rightmost digits. The stems are listed individually, with all leaf values corresponding to each stem displayed beside that stem. A pie chart is a circular depiction of data. The amount of each category is represented as a slice of the pie proportionate to the total. The researcher is cautioned in using pie charts because it is sometimes difficult to differentiate the relative sizes of the slices. The bar chart or bar graph uses bars to represent the frequencies of various categories. The bar chart can be displayed horizontally or vertically. A Pareto chart is a vertical bar chart that is used in total quality management to graphically display the causes of problems. The Pareto chart presents problem causes in descending order to assist the decision maker in prioritizing problem causes. The scatter plot is a two-dimensional plot of pairs of points from two numerical variables. It is used to graphically determine whether any apparent relationship exists between the two variables.

2.1 Sources of data

i) Primary data ii) Secondary data

Primary Data: The data are collected through first hand investigation without any internal records or any published sources are called primary data.

Example: Survey data, interview data, questionnaire data etc.

Uses of Primary Data

The common uses of primary data are mentioned below:

- (1) They are used as the basis of measurements.
- (2) They are used for original study.
- (3) They are direct investigations.
- (4) They are used for very recent information.

They are used for reducing uncertainty

2.2 Methods/Sources of Collecting Primary Data

Primary data can be collected by the following sources:

- (1) Direct investigation,
- (2) Direct interviews,
- (3) Telephone interviews,
- (4) Filling questionnaire,
- (5) Personal observations,
- (6) Counting by enumerators,

- (7) Mailing,
- (8) Surveys,
- (9) Census, etc.

Statistical data as we have seen can be either primary or secondary. Primary data are those which are collected for the first time and so are in crude form. But secondary data are those which have already been collected.

Primary data are always collected from the source. It is collected either by the investigator himself or through his agents. There are different methods of collecting primary data. Each method has its relative merits and demerits. The investigator has to choose a particular method to collect the information. The choice to a large extent depends on the preliminaries to data collection some of the commonly used methods are discussed below.

1. Direct Personal observation:

This is a very general method of collecting primary data. Here the investigator directly contacts the informants, solicits their cooperation and enumerates the data. The information are collected by direct personal interviews.

The novelty of this method is its simplicity. It is neither difficult for the enumerator nor the informants. Because both are present at the spot of data collection. This method provides most accurate information as the investigator collects them personally. But as the investigator alone is involved in the process, his personal bias may influence the accuracy of the data. So it is necessary that the investigator should be honest, unbiased and experienced. In such cases the data collected may be fairly accurate. However, the method is quite costly and time-consuming. So the method should be used when the scope of enquiry is small.

2. Indirect Oral Interviews:

This is an indirect method of collecting primary data. Here information are not collected directly from the source but by interviewing persons closely related with the problem. This method is applied to apprehend culprits in case of theft, murder etc. The informations relating to one's personal life or which the informant hesitates to reveal are better collected by this method. Here the investigator prepares 'a small list of questions relating to the enquiry. The answers (information) are collected by interviewing persons well connected with the incident. The investigator should cross-examine the informants to get correct information.

This method is time saving and involves relatively less cost. The accuracy of the information largely depends upon the integrity of the investigator. It is desirable that the investigator should be experienced and capable enough to inspire and create confidence in the informant to collect accurate data.

3. Mailed Questionnaire method:

This is a very commonly used method of collecting primary data. Here information are collected through a set of questionnaire. A questionnaire is a document prepared by the investigator containing a set of questions. These questions relate to the problem of enquiry directly or indirectly. Here first the questionnaires are mailed to the informants with a formal request to answer the question and send them back. For better response the investigator should bear the postal charges. The questionnaire should carry a polite note explaining the aims and objective of the enquiry, definition of various terms and concepts used there. Besides this the investigator should ensure the secrecy of the information as well as the name of the informants, if required.

Success of this method greatly depends upon the way in which the questionnaire is drafted. So the investigator must be very careful while framing the questions. The questions should be

- (i) Short and clear
- (ii) Few in number
- (iii) Simple and intelligible
- (iv) Corroboratory in nature or there should be provision for cross check
- (v) Impersonal, non-aggressive type
- (vi) Simple alternative, multiple-choice or open-end type

(a) In the simple alternative question type, the respondent has to choose between alternatives such as 'Yes or No', 'right or wrong' etc.

For example: Is Adam Smith called father of Statistics? Yes/No,

(b) In the multiple choice type, the respondent has to answer from any of the given alternatives.

Example: To which sector do you belong?

- (i) Primary Sector
- (ii) Secondary Sector
- (iii) Tertiary or Service Sector

(c) In the Open-end or free answer questions the respondents are given complete freedom in answering the questions. The questions are like - What are the defects of our educational system?

The questionnaire method is very economical in terms of time, energy and money. The method is widely used when the scope of enquiry is large. Data collected by this method are not affected by the personal bias of the investigator. However the accuracy of the information depends on the cooperation and honesty of the informants. This method can be used only if the informants are cooperative, conscious and educated. This limits the scope of the method.

4. Schedule Method:

In case the informants are largely uneducated and non-responsive data cannot be collected by the mailed questionnaire method. In such cases, schedule method is used to collect data. Here the questionnaires are sent through the enumerators to collect information. Enumerators are persons appointed by the investigator for the purpose. They directly meet the informants with the questionnaire. They explain the scope and objective of the enquiry to the informants and solicit their cooperation. The enumerators ask the questions to the informants and record their answers in the questionnaire and compile them. The success of this method depends on the sincerity and efficiency of the enumerators. So the enumerator should be sweet-tempered, good-natured, trained and well-behaved.

Schedule method is widely used in extensive studies. It gives fairly correct result as the enumerators directly collect the information. The accuracy of the information depends upon the honesty of the enumerators. They should be unbiased. This method is relatively more costly and time-consuming than the mailed questionnaire method.

5. Local Agents:

Sometimes primary data are collected from local agents or correspondents. These agents are appointed by the sponsoring authorities. They are well conversant with the local conditions like language, communication, food habits, traditions etc. Being on the spot and well acquainted with the nature of the enquiry they are capable of furnishing reliable information.

The accuracy of the data collected by this method depends on the honesty and sincerity of the agents. Because they actually collect the information from the spot. Information from a wide area at less cost and time can be collected by this method. The method is generally used by government agencies, newspapers, periodicals etc. to collect data.

Information is like raw materials or inputs in an enquiry. The result of the enquiry basically depends on the type of information used. Primary data can be collected by employing any of the above methods. The investigator should make a rational choice of the methods to be used for collecting data. Because collection of data forms the beginning of the statistical enquiry.

2.3 Secondary Data: The data are collected from any medium which has already been collected by others are called secondary data.

Example: Journal, thesis, published statistics etc.

Uses of Secondary Data

The common uses of secondary data are mentioned below:

- (1) They are used for decisions making.
- (2) They are suitable for multipurpose uses.
- (3) They are used for reference purposes.
- (4) They are used for documentary project.

- (5) They are used for minimizing costs.

Methods/Sources of Collecting Secondary Data

Secondary data can be collected by the following sources:

- (1) Journals,
- (2) Reports,
- (3) Institutional publications,
- (4) Public library,
- (5) Information bureau,
- (6) Govt. agencies,
- (7) International agencies (UNO, FAO, WHO etc),
- (8) National and international news media,
- (9) Statistical books, etc.

Secondary data are second hand informations. They are not collected from the source as the primary data. In other words, secondary data are those which have already been collected. So they may be relatively less accurate than the primary data. Secondary data are generally used when the time of enquiry is short and the accuracy of the enquiry can be compromised to some extent. Secondary data can be collected from a number of sources which can broadly be classified into two categories.

- i) Published sources
- ii) Unpublished sources

Published Sources:

Mostly secondary data are collected from published sources. Some important sources of published data are the following.

1. Published reports of Central and State Governments and local bodies.
2. Statistical abstracts, census reports and other reports published by different ministries of the Government.
3. Official publications of the foreign Governments.
4. Reports and Publications of trade associations, chambers of commerce, financial institutions etc.
5. Journals, Magazines and periodicals.
6. Periodic Publications of Government organizations like Central Statistical Organization (C. S. O.), National Sample Survey Organization (NSSO).
7. Reports submitted by Economists, Research Scholars, Bureaus etc.
8. Published works of research institutions and Universities etc.

Unpublished Sources:

Statistical data can also be collected from various unpublished sources. Some of the important unpublished sources from which secondary data can be collected are:

1. The research works carried out by scholars, teachers and professionals.
2. The records maintained by private firms and business enterprises. They may not like to publish the information considering them as business secret.
3. Records and statistics maintained by various departments and offices of the Central and State Governments, Corporations, Undertakings etc.

Secondary data are already collected information. They might have been collected for some specific purposes. So they must be used with caution. It is generally very difficult to verify such information to find out inconsistencies, errors, omissions etc. Therefore scrutiny of secondary data is essential. Because the data might be inaccurate, unsuitable or inadequate. Thus it is very risky to use statistics collected by other people unless they have been thoroughly edited and found reliable, adequate and suitable for the purpose.

2.3.1 Advantages and Disadvantages of Secondary data Advantages

Disadvantages:

1. Faster
2. Less Expensive
3. Less activities (Field trip, Survey etc.)

Disadvantages:

1. Not easily available
2. Not adequate
3. May not meet the needs of researcher
4. Outdated information
5. Variation in definition
6. Inaccurate or bias

2.4 Properties of a Good Questionnaire

A good questionnaire has the following properties:

- (1) It should be easier to understand.
- (2) It should be clear to express the facts.
- (3) It should be concise form.
- (4) The number of questions should be as few as possible.
- (5) Questions should be logically arranged.

- (6) Questions should be short and simple.
- (7) Questions of a sensitive nature should be avoided.
- (8) Instructions should have to the information.
- (9) It should be just in time.
- (10) Questions should be capable of objective answer i.e., Yes or No.

2.8 Distinctions between primary data and secondary data

Primary Data	Secondary data
(1) It is first hand conducted data.	(1) It is second level collected data.
(2) It is direct data.	(2) It is indirect data.
(3) It may be unprocessed.	(3) It is always processed.
(4) It is collect from the origin of source.	(4) It is collected through media.
(5) It is expensive.	(5) It is not expensive.
(6) It takes huge time.	(6) It does not take huge time.
(7) It creates secondary data	(7) It creates decisions.
(8) It gives very recent information.	(8) It gives late information.
(9) It suitable for researchers.	(9) It suitable for analyzers.
(10) It is suitable for original study.	(10) It is suitable for relative study.

Example : The population consisting of all possible outcomes (here head and tail) in successive tosses of a coin.

Sample : A representative and considerably small part of a population is known as a sample of that population.

Example :

(i) A group of 1050 students from 31,000 students of Dhaka University constitute a sample.

(ii) A spoonful boiling rice from a cooking pot of boiling rice.

2.02 : Variable

The characteristic which varies over the units (or from unit to unit) is called variable.

Example : Father's occupation of the students of Dhaka University. Here, the unit or population-unit is a student and their father's occupation varies. Say, one father's occupation is teaching, another father's occupation is business, else other father's occupation is driving and so on.

2.03 : Classification of variable

We can classify the variables according to the characteristics into two groups :

(a) Qualitative variable

(b) Quantitative variable

(a) Qualitative variable : There are many characteristics that can not be expressed in any numerical form (i. g., color, sex, occupation etc) but we can arrange them according to their quality or attribute. These types of characteristic are known as qualitative variable.

Example : Father's occupation of the students of Dhaka University.

(b) Quantitative variable : The characteristics of an unit or item that are expressed in numerical form or in numbers are called quantitative variable.

Example : Monthly family income of the students of Dhaka University. Quantitative variable can further be classified into two sub-groups :

(i) Discrete variable

(ii) Continuous variable

(i) Discrete variable

A quantitative variable which possesses isolated or integral value is called discrete variable.

Example : Family size, population size. Number of road accidents per day in Bangladesh.

Population : Simply, population means the aggregate of human individuals in a defined area or region. In statistics, population refers the totality of all the items or individuals having some specific characteristics. e.g., All the students of university of Dhaka constitute a population.

A population can be classified into one of two groups-

(a) **Finite population :** A population having a finite number of units (or individuals or items) is called a finite population.

Example : The population consisting the students of Dhaka University.

(b) **Infinite population :** A population having an infinite number of units (or individuals or items) is called an infinite population.